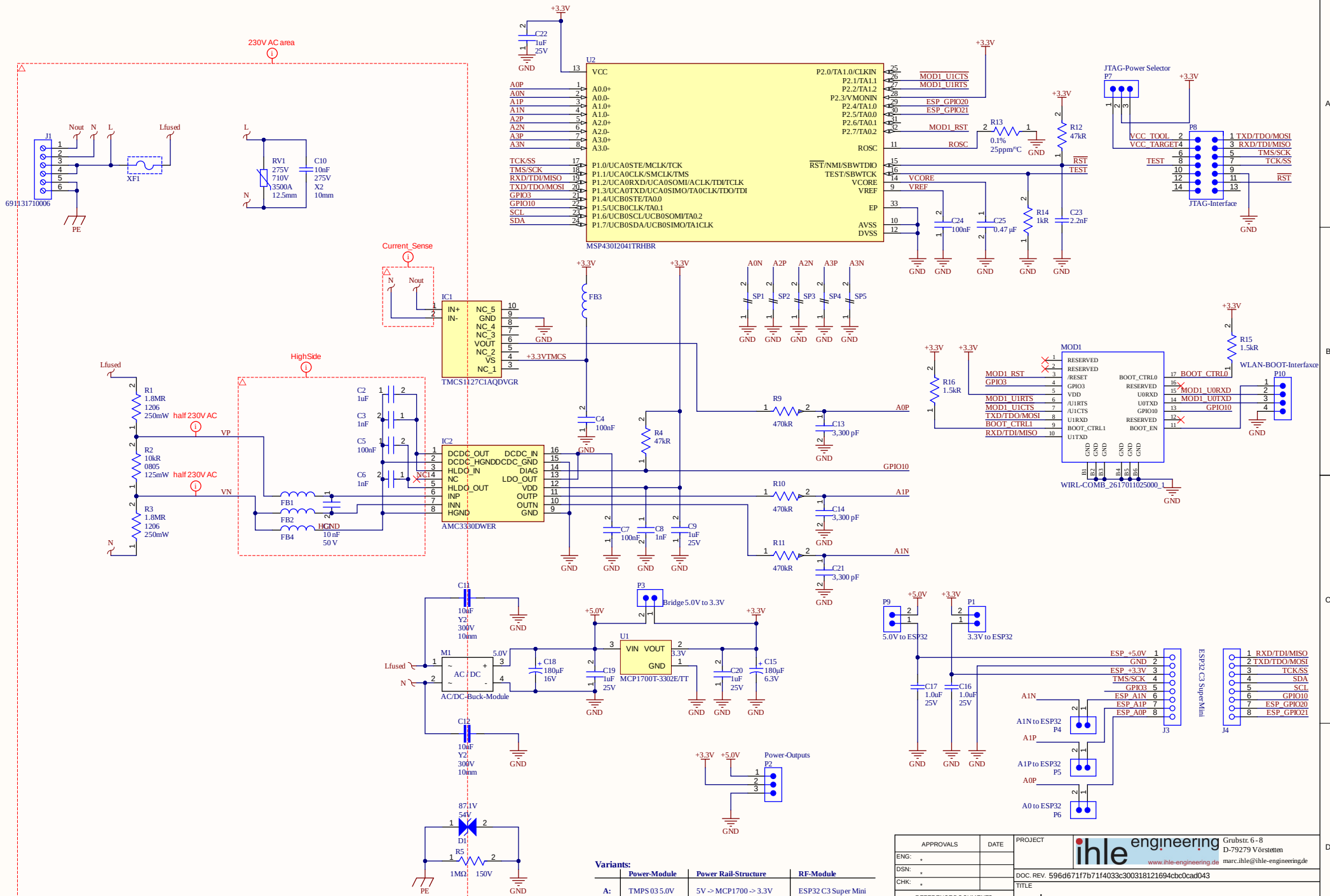




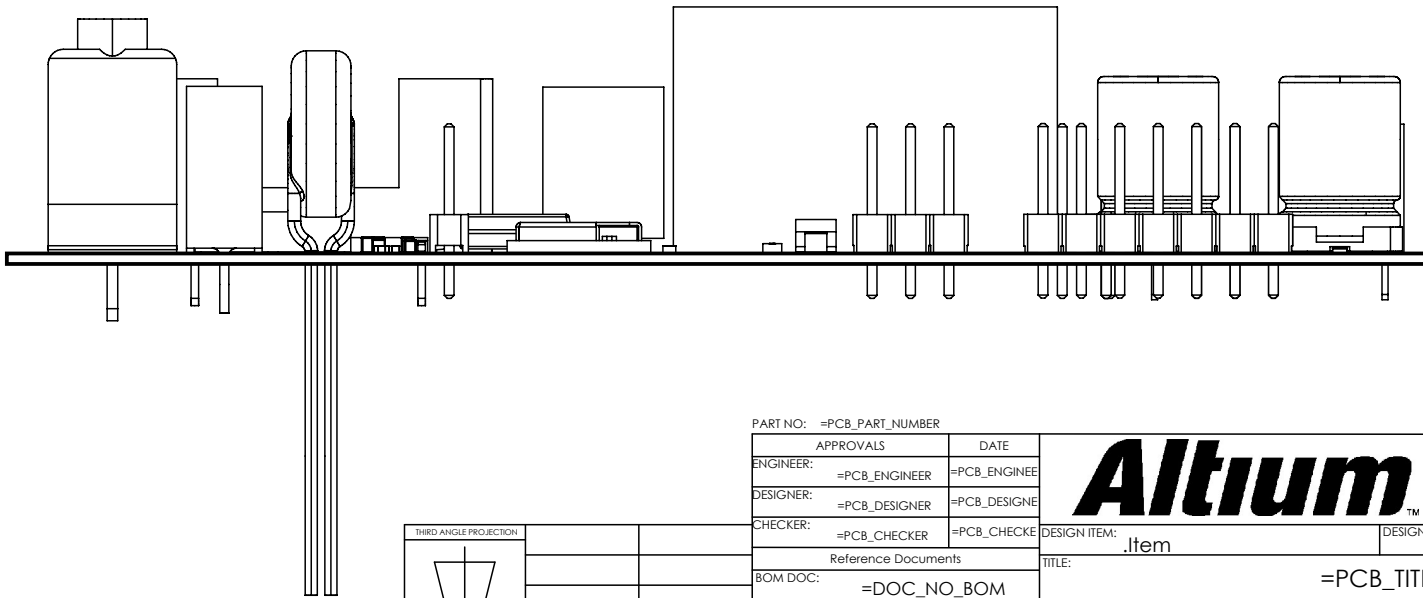
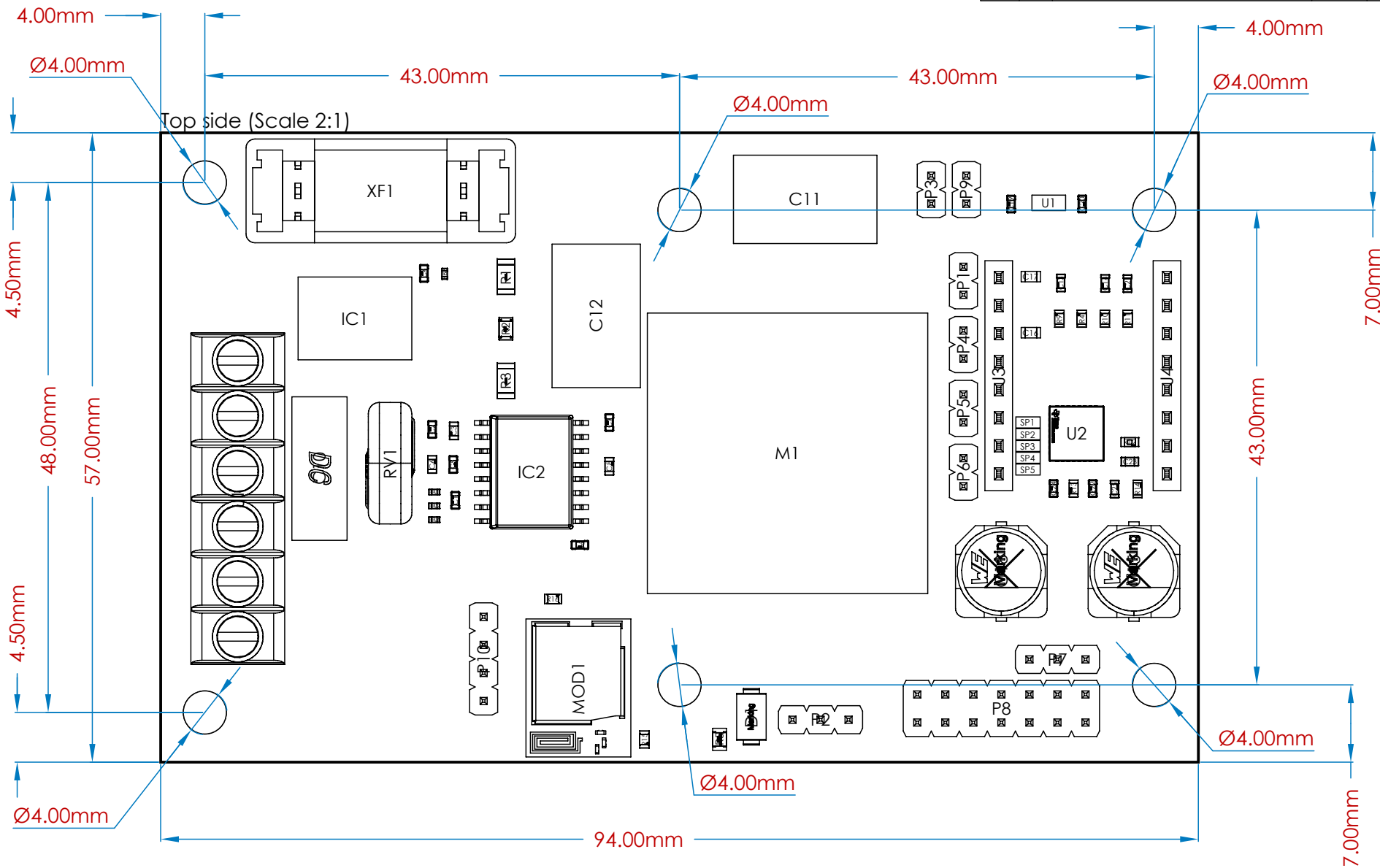
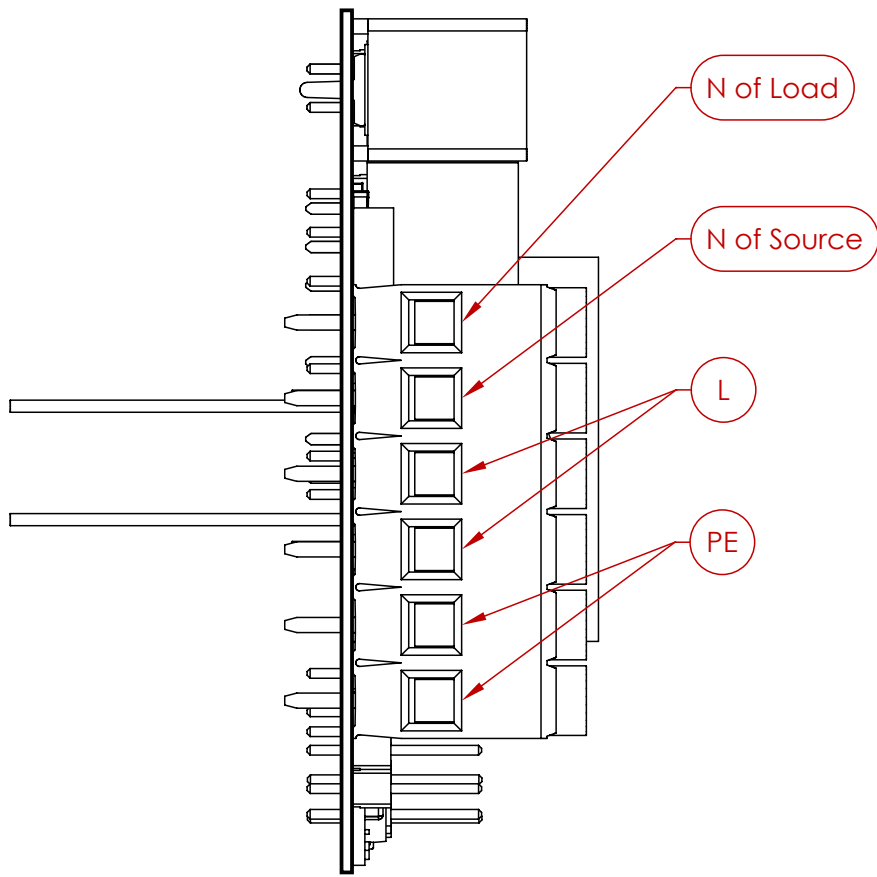
Variants:

	Power-Module	Power Rail-Structure	RF-Module
A:	TMPS03 5.0V	5V → MCP1700 → 3.3V	ESP32 C3 Super Mini
B:	AM11-12W05V	5V → MCP1700 → 3.3V	ESP32 C3 Super Mini
C:	TMPS03 3.3V	5V = 3.3V (P3 set)	Stephano-I

APPROVALS	DATE	PROJECT	ihle engineering	Grubstr. 6-8 D-79279 Vörsstetten
ENG: *		DOC. REV. 596d671f7b71f4033c300318121694cb0cad043	www.ihle-engineering.de	marc.ihle@ihle-engineering.de
DSN: *		TITLE		
CHK: *		★		
BOM:		REFERENCE DOCUMENTS		
ASSY DWG:		SIZE	DWG NO.	REV
FAB DWG:		A3		
PCB DWG:		FILE NAME	AC-Power-Monitor.SchDoc	SHEET 1 OF 1

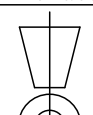
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REV STATUS OF SHEETS	REV	-	-	-	-	-	-	-	-	-	REVISIONS				
	SHEET	-	-	-	-	-	-	-	-	-	ZONE	REV	DESCRIPTION	DATE	APPROVED
											--	-	-----	--/--/--	-, -----
											--	-	-----	--/--/--	-, -----
											--	-	-----	--/--/--	-, -----
											--	-	-----	--/--/--	-, -----
											--	-	-----	--/--/--	-, -----



NOTES UNLESS OTHERWISE SPCEIFIED:

1. Board shape optimized for BOPLA SE 435 DE - 9016 housing.
2. NOTE
3. NOTE

		PART NO: =PCB_PART_NUMBER		Altium™										
		APPROVALS										DATE		
		ENGINEER: =PCB_ENGINEER										=PCB_ENGINEER		
		DESIGNER: =PCB_DESIGNER										=PCB_DESIGNER		
THIRD ANGLE PROJECTION				CHECKER: =PCB_CHECKER	=PCB_CHECKER		DESIGN ITEM: .Item			DESIGN ITEM REVISION: .ItemRevision				
				Reference Documents		TITLE:								
				BOM DOC: =DOC_NO_BOM		=PCB_TITLE_1								
				ASSY DOC: =DOC_NO_FAB_DWG		=DRAF =PCB_TITLE_2								
				SCH DOC: =DOC_NO_SCH_DWG		SIZE: A3		CAGE CODE: =CAGE_CO		DWG NO: =DOC_NO_ASSY_DWG			REV:	
NEXT ASSY		USED ON		PCB DOC: =PCB_DWG_NO		SCALE:		FILE NAME: New		AC-Power-Monitor.PCBdwf			SHEET: 1 OF 5	
		APPLICATION												

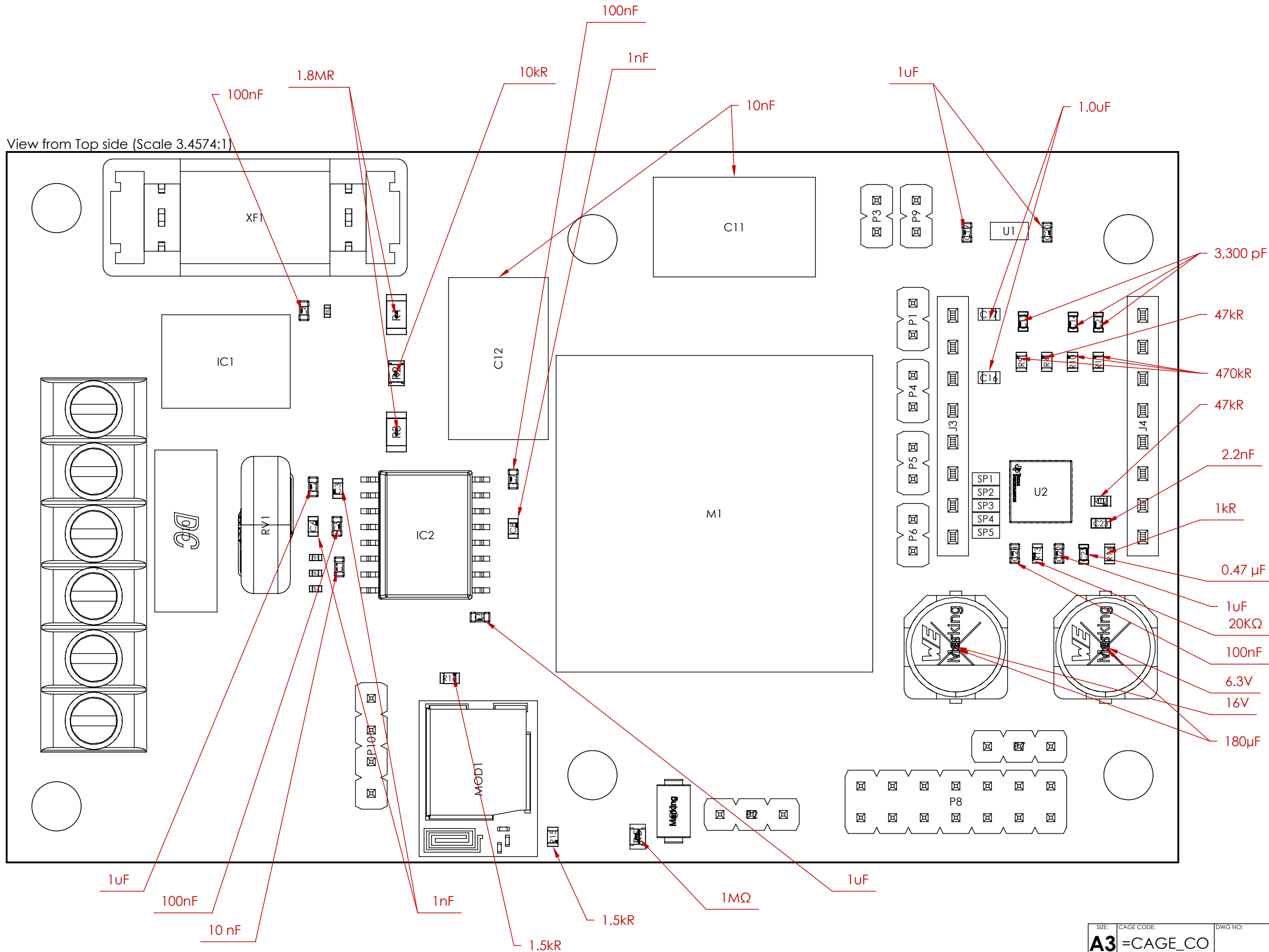
Bill Of Materials

Line #	Designator	Name	Quantity	Purpose
	C1	CC0603KRX7R9BB103	1	
	C2, C9, C19, C20, C22	CGA3E1X7R1E105K080AC	5	
	C3, C6, C8	06035C102KAT2A	3	
	C4, C5, C7, C24	CGA3E2X7R1H104K080AA	4	
	C10	10nF	1	
	C11, C12	MKY22W21003I00	2	
	C13, C14, C21	C0603C332J5GACTU	3	
	C15	180µF	1	
	C16, C17	C0603C105K3RACTU	2	
	C18	180µF	1	
	C23	885012206085	1	
	C25	C0603C474K4RACTU	1	
	D1	400W, 54VDC	1	
	FB1, FB2, FB3, FB4	1800Ohm	4	
	IC1	TMCS1127C1AQDVGR	1	
	IC2	AMC3330DWER	1	
	J1	691131710006	1	
	J3, J4	61300811821	2	
	M1	AC/DC-Buck-Module	1	
	MOD1	WIRL-COMB_2617011025000_1	1	
	P1, P3, P4, P5, P6, P9	61300211121	6	3.3V to ESP32, Bridge 5.0V to 3.3V, A1N to ESP32, A1P to ESP32, A0 to ESP32, 5.0V to ESP32
	P2, P7	61300311121	2	Power-Outputs, JTAG- Power Selector
	P8	61301421121	1	JTAG-Interface
	P10	61300411121	1	WLAN-BOOT- Interfaxce
	R1, R3	RK73H2BTTD1804F	2	
	R2	CRCW080510K0FKEA	1	
	R4	CRCW060347K0FKEA	1	
	R5	CRGCQ0805F1M0	1	
	R9, R10, R11	CRCW0603470KFKEA	3	
	R12	KTR03EZPF4702	1	
	R13	CPF0603B20KE1	1	
	R14	AC0603FR-071KL	1	
	R15, R16	CRCW06031K50FKEA	2	
	RV1	275VAC, 350VDC	1	
	U1	MCP1700T-3302E/TT	1	
	U2	MSP430I2041TRHBR	1	
	XF1	696106003002	1	

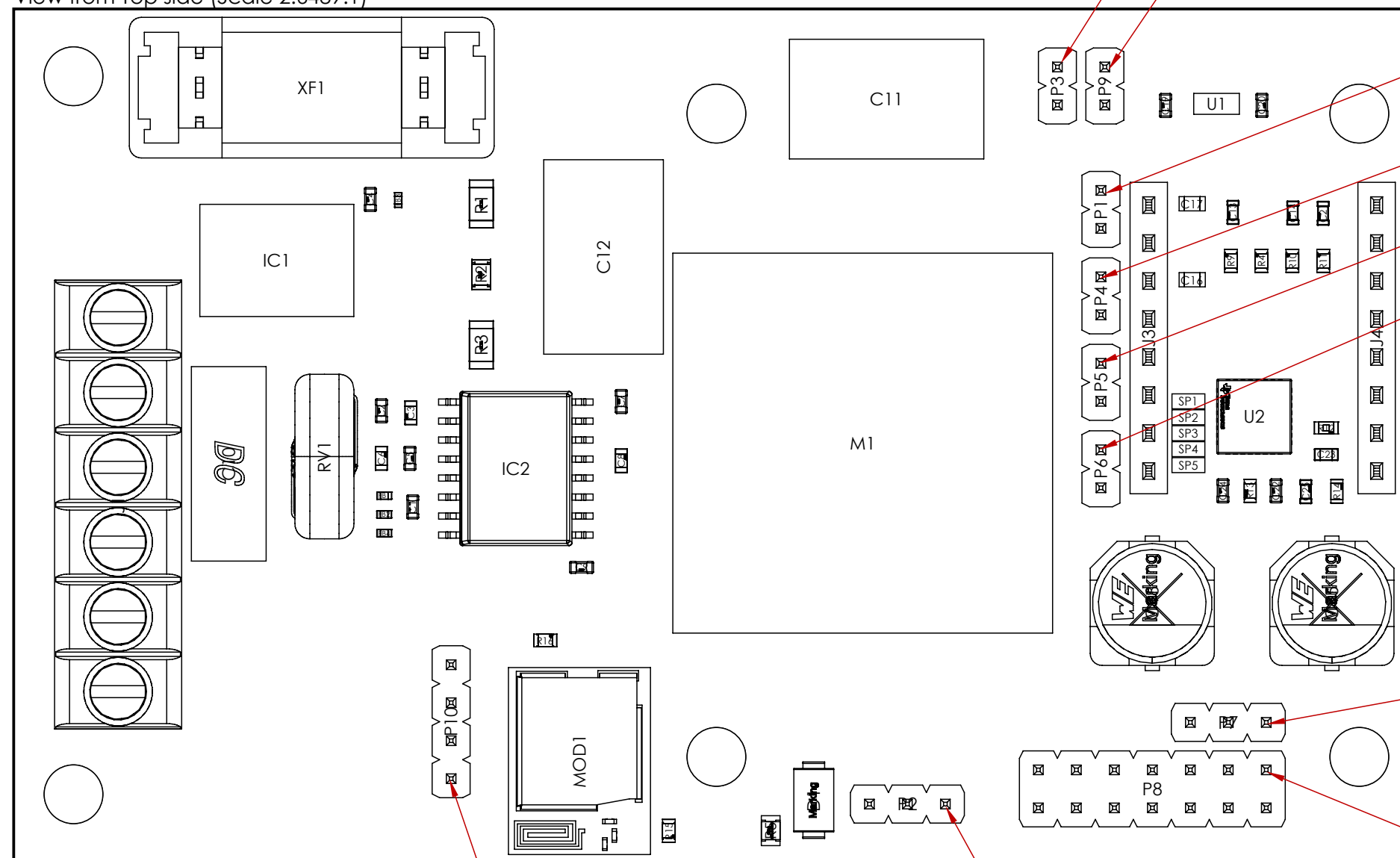
Layer Stack Legend

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.01mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.04mm		Signal	GTL
Prepreg		0.07mm	PP-006	Dielectric	
CF-004	N / GND	0.07mm		Signal	G1
		0.32mm	FR-4	Dielectric	
CF-004	L / VCC	0.07mm		Signal	G2
Prepreg		0.07mm	PP-006	Dielectric	
Copper	Bottom Layer	0.04mm		Signal	GBL
Surface Material	Bottom Solder	0.01mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

Total thickness: 0.69mm



View from Top side (Scale 2.6489:1)



Jumper: Bridge 5.0V-Net to 3.3V-Net
Don't bridge if a 5.0V DC/DC-Converter is in use!

Jumper: Connect 5.0V to ESP32

Jumper: Connect 3.3V to ESP32

Jumper: Connect A1N to ESP32

Jumper: Connect A1P to ESP32

Jumper: Connect A0P to ESP32

Jumper:
right: VCC Tool
left: VCC Target

Pin1, JTAG-Interface

MOD1-Debugger

5.0V, 3.3V, GND



Bill of Materials for AC-Power-Monitor.PrjPcb

Source Data From: AC-Power-Monitor.BomDoc
Variant: None
Report Date: 20.04.2025 21:20

Primary										Secondary				
Altium Vault PIN	Designator	Description	Qty	Supplier	Supplier Part Number	Manufacturer	Manufacturer Part Number	Supplier Unit Price	Supplier	Supplier Part Number	Manufacturer	Manufacturer Part Number	Supplier Unit Price	
CMP-2100-03600-2	C1	CAP CER 10000PF 50V X7R 0603	1	Mouser	603-CC603KRX7R9BB103	Yageo	CC603KRX7R9BB103	0.1003						
CMP-08246-001507-1	C2, C22	Cap Ceramic 1uF 25V X7R 10% Pad SMD 0603 +125°C Automotive TIR	2	Mouser	810-CGA3E1X7R1E105K	TDK	CGA3E1X7R1E105K080AC	0.1605						
CMP-2006-04378-2	C3, C6	General Purpose Ceramic Capacitor, 0603, 1nF, 10%, X7R, 15%, 50V	2	Mouser	581-06035C102K	Kyocera AVX	06035C102KA72A	0.0211						
CMP-2000-07546-2	C4, C7	Cap Ceramic 100nF 50V X7R 10% Pad SMD 0603 +125°C Automotive TIR	2	Mouser	810-CGA3E2X7R1H104K	TDK	CGA3E2X7R1H104K080AA	0.1003						
CMP-2000-07546-2	C5	Cap Ceramic 100nF 50V X7R 10% Pad SMD 0603 +125°C Automotive TIR	1	Mouser	810-CGA3E2X7R1H104K	TDK	CGA3E2X7R1H104K080AA	0.1003						
CMP-2006-04378-2	C8	General Purpose Ceramic Capacitor, 0603, 1nF, 10%, X7R, 15%, 50V	1	Mouser	581-06035C102K	Kyocera AVX	06035C102KA72A	0.0211						
CMP-08246-001507-1	C9, C19, C20	Cap Ceramic 1uF 25V X7R 10% Pad SMD 0603 +125°C Automotive TIR	3	Mouser	810-CGA3E1X7R1E105K	TDK	CGA3E1X7R1E105K080AC	0.1605						
890324023006CS	C10	WCAP-F1X2 Film Capacitor, THF, L13 x W5 x H1mm, 10nF, 275V	1	Würth	890324023006CS	Würth	890324023006CS	0.4079						
C_unipolar	C11, C12		2	Mouser	505-MKY22W2100300KSS	WIMA	MKY22W2100300KSSD	0.8327						
CMP-2006-03334-2	C13, C14, C21	CAP CER 3300PF 50V 5% NPO 0603	3	Mouser	80-C0603C332J5G	KEMET	C0603C332J5GACTU	0.2809						
875075155001, 875075355001	C15, C18	WCAP-PSLC Aluminum Polymer Capacitor, V-Chip SMT, D8 x H11.7mm, 180uF, 6.3V, WCAP-PSLC Aluminum Polymer Capacitor, V-Chip SMT, D8 x H11.7mm, 180uF, 16V	2	[NoParam], Digipley	[NoParam], 732-6488-1-ND	Würth Electronics	875075155001, 875075355001							
CMP-2006-03713-1	C16, C17		2	Mouser	80-C0603C105K3R	KEMET	C0603C105K3RACTU	0.2508						
CMP-03913-53914794-1	C23	885012206085 Multilayer Ceramic Capacitor, Wcap-Csgp Series, 2200 Pf, 10%, X7R, 50 V, 0603 [1608 Metric] Rohs Compliant: Yes...	1	Mouser	710-885012206085	Würth Electronics	885012206085	0.1003						
CMP-2000-07546-2	C24	Cap Ceramic 100nF 50V X7R 10% Pad SMD 0603 +125°C Automotive TIR	1	Mouser	810-CGA3E2X7R1H104K	TDK	CGA3E2X7R1H104K080AA	0.1003						
CMP-2006-03236-2	C25	CAP CER 0.47UF 16V X7R 0603	1	Mouser	80-C0603C474K4R	KEMET	C0603C474K4RACTU	0.1204						
824501541	D1	WE-TVSP SMT Power TVS Diode, DO-214AC, Bidirectional, 400W, 54VDC	1	Mouser	710-824501541	Würth Electronics	824501541	0.2909						
74269244182	FB1, FB2, FB3, FB4	WE-TMSB Tiny Multilayer Suppression Bead, size 0402, 1800Ohm, 0.21A	4	Digipkey	732-74269244182CT-ND	Würth Electronics	74269244182	0.1104						
TMCS1127CIAQDVGR	IC1	Integrated Circuit	1			Texas Instruments	TMCS1127CIAQDVGR							
AMC3330DWER	IC2	Integrated Circuit	1	Mouser	595-AMC3330DWER	Texas Instruments	AMC3330DWER	6.4900						
691131710006	J1	WR-TBL Series 1317 Horizontal Entry, THF, pitch 5mm, 06p	1	Mouser	710-691131710006	Würth Electronics	691131710006	4.1000						
61300811821	J3, J4	WR-PHD Socket Header, THF, Vertical, pitch 2.54mm, 1 Row, 8P	2	Mouser	710-61300811821	Würth Electronics	61300811821	0.4214						
ACDC-Buck-Module	M1		1	Arrow Electronics	TMPS 03-105	Traco Power	TMPS03-105	17.4600						
WRL-COMB_2617011025000_1	MOD1	Stephano-I - Combined 2.4 GHz Wi-Fi and Bluetooth® LE 5.0 Module	1	Digipkey	732-2617011025000DKR-ND	Würth Elektronik	2617011025000	10.6300						
61300211121	P1, P3, P4, P5, P6, P9	WR-PHD Pin Header, THF, Vertical, pitch 2.54mm, 1 Row, 2P	6	Mouser	710-61300211121	Würth Electronics	61300211121	0.1304						
61300311121	P2, P7	WR-PHD Pin Header, THF, Vertical, pitch 2.54mm, 1 Row, 3P	2	Mouser	710-61300311121	Würth Electronics	61300311121	0.1304						
61301421121	P8	WR-PHD Pin Header, THF, Vertical, pitch 2.54mm, 2 Row, 14P	1	Mouser	710-61301421121	Würth Electronics	61301421121	0.9631						
61300411121	P10	WR-PHD Pin Header, THF, Vertical, pitch 2.54mm, 1 Row, 4P	1	Mouser	710-61300411121	Würth Electronics	61300411121	0.1906						
CMP-2003-01283-2	R1, R3	1.8MΩ ±1% 0.25W Chip Resistor 1206 Thick Film AEC-Q200 Qualified	2	Mouser	660-RK73H2BTTD1804F	KOA Speer	RK73H2BTTD1804F	0.1003						
CMP-2001-04908-2	R2	RES Thick Film, 10kΩ, 1%, 0.125W, 100ppm/°C, 0805	1	Newark	68AH8357	Vishay	CRCW080510K0FKEA	0.0261						
CMP-2000-03379-3	R4	RES SMD 47K OHM 1% 1/10W 0603	1	Mouser	71-CRCW0603-47K-E3	Vishay	CRCW060347K0FKEA	0.1003						
CMP-03211-002463-1	R5	CRGCO0805 1MΩ 0.125W ±100ppm/°C 1% 150V	1	Newark	73AC2562	TE Connectivity	CRGCO0805F1M0	0.0040						
CMP-2000-03370-2	R9, R10, R11	RES Thick Film, 470kΩ, 1%, 0.1W, 100ppm/°C, 0603	3	Newark	52K8555	Vishay	CRCW060347K0FKEA	0.0191						
CMP-08839-030818-1	R12	Res Thick Film 0603 47kΩ 1% 0.1W ±100ppm/°C Automotive Paper TIR	1	Mouser	755-KTR03E2PF4702	Rohm	KTR03E2PF4702	0.1605						
CMP-03211-006634-1	R13	RES SMD 20KΩ 0.1% 1/16W 0603	1	Arrow Electronics	CPF0603B20KE1	TE Connectivity	CPF0603B20KE1	0.1858						
CMP-2000-04828-3	R14	RES 1kΩ 1.00% 0.1W 0603	1	Digipkey	311-1KLDOKR-ND	Yageo	AC0603R-071KL	0.1003						
CMP-2000-02995-2	R15, R16	RES Thick Film, 1.5kΩ, 1%, 0.1W, 100ppm/°C, 0603	2	Avnet	61M5331	Vishay	CRCW06031K50FKEA	0.0161						
820412711	RV1	WE-VD Disk Varistor, THF, 275VAC, 350VDC, 190pF	1	Mouser	710-820412711	Würth Electronics	820412711	0.5718						
CMP-0136-00788-1	U1	Low Quiescent Current LDO, 3-Pin SOT-23, Extended Temperature, Tape and Reel	1	Arrow Electronics	MCP1700T-3002EFT	Microchip	MCP1700T-3002EFT	0.4473						
CMP-04869-000074-1	U2	IC MCU 16BIT 32KB FLASH 32VQFN	1	Mouser	595-MSP43020411TR-BR	Texas Instruments	MSP43020411TR-BR	3.1200						
696106003002	XF1	WR-FSH Fuseholder THF Blocks PCB Clip Cover	1	Mouser	710-696106003002	Würth Electronics	696106003002	0.5819						